

Meeting Minutes: Webinar on DCAT-AP (SEMIC - 04.02)

Project:	SEMIC	Date and Time:	04/02/2025 10:00 - 12:00
Meeting Type:	Webinar	Location:	Virtual
Coordinators:	Pavlina Fragkou Emiel Dhondt	Issue Date:	05/03/2025

Agenda of the webinar		
13:00 - 13:05	Introduction	Slides 2-9
13:05 - 13:10	Codelists	Slides 10 - 37
13:10 - 13:30	Issues with GeoDCAT-AP alignment	Slides 38 - 42
13:30 - 13:50	Issues with a feedback request	Slides 43 - 51
13:50 - 14:00	Next steps	Slides 53

Meeting Slides
LINK

Participants		
Name	Initials	Organisation
Ahmed Abid	AA	NTT Data
Alex Akkila Forsberg	AAK	Not Shared

Participants		
Name	Initials	Organisation
Ana Velasco	AV	IGN Spain
Anastasia Sofou	AS	SEMIC Team
Andras	A	Not Shared
Andreas Fogelberg	AF	Not Shared
Anja Loddenkemper	AL	GDI-NI
Anna Miglio	AM	Royal Observatory of Belgium
Annie Royen	AR	Not Shared
António Soares	AS	AMA, Portugal
Bart Hanssens	BH	BOSA, Belgium
Bert Van Nuffelen	BVN	SEMIC Team
Berta	B	CR
Eva Christina Andersson	ECA	Not Shared
Casper le Gras	CLG	Not Shared
Christian Wittig	CW	SK Hamburg
Costas Simatos	CS	Not Shared
Daniela Witter	DW	Not Shared
Darko	D	Not Shared
Emidio Stani	ES	SEMIC Team
Emiel Dhondt	ED	SEMIC Team

Participants		
Name	Initials	Organisation
Enric Staromiejski	ES	NTT Data
Eva Thelisson	ET	Not Shared
Faruk Karabulut	FK	SEMIC Team
Flojere Thaqi Blakaj	FTB	Not Shared
Floor Van Damme	FVD	Not Shared
George Mourkousis	GM	HSE
Gérald	G	Not Shared
Haluk Inanmis	HI	Inted Ltd
Ine Weyts	IW	SEMIC Team
Israel Romero	IR	Not Shared
Jakub Klimek	JK	SEMIC Team
Jan Meijer	JM	Not Shared
Jan R.	JR	Not Shared
Jean-Christophe Haffner	JCH	SEMIC Team
Jens Kuscherka	JK	Not Shared
Jim Yang	JY	NO
Jan Hjelaenger	JH	DNK
Joachim Nielandt	JN	BE
Jordi Escriu	JE	EC-JRC

Participants		
Name	Initials	Organisation
José Pedro	JP	Not Shared
Juliette Legrand	JL	Not Shared
Kees Trautwein	KT	Architect GU
Kuldar Aas	KA	EE
Lars	L	DK
Lars-Inge Arnevik	LIA	NO
Laura	L	Not Shared
Peter Lubrich	PL	BASt
Ludger Rinsche	LR	Not Shared
Agata Majchrowska	AM	Not Shared
Manuel Jañez	MJ	ES-datos.gob.es
Mariela Rodriguez	MR	Not Shared
M.Adam	MA	Not Shared
Martin Alvarez	MA	Not Shared
Matthias Grönewald	MG	FITKO
Matthias Palmer	MP	Not Shared
Miguel Cea	MC	Not Shared
Josema Alonso	JA	SEMIC Team
Szilvia Mészáros	SM	Not Shared

Participants		
Name	Initials	Organisation
Michal Kuban	MK	CNECT
Mohamed Chouaiech	MC	DG SANTE
Niels Hoffmann	NH	NL / Geonovum
Natalia Zylinska Puta	NZP	DG SANTE
Nataliia	N	Not Shared
De Ville	DV	RBINS
Oceane Abrard	OA	Not Shared
Øystein Åsnes	ØÅ	NO
Pablo Diez	PD	Spanish Ministry of Digital Transformation
Paloma Abad	PA	Not Shared
Pavlina Fragkou	PF	DIGIT
Peter Bruhn Andersen	PBA	Danish Agency for Digital Goverment
Pedro Polonia	PP	EDA
Pierre Billot	PB	Not Shared
Priyanka	P	Not Shared
RMIS	R	Not Shared
Antonio Rotundo	AR	IT
Sander Van Dooren	SVD	DIGIT
Simon Dutkowski	SD	FOKUS

Participants		
Name	Initials	Organisation
Stijn Tallir	ST	Not Shared
Svetlogor Kirov	SK	Not Shared
Terje Sylvarnes	TS	The Norwegian Digitalisation Agency
Thomas Tursics	TT	DE, GovData
Uwe Voges	UW	Not Shared
Elizabeth	E	Not Shared
Sebastian Sklarß	SS	Not Shared

Points discussed and decisions taken

Topics discussed	Outcome
Codelists	
Issue #407	Approved
Issue #408	Approved
Issue #409	Issue raised to GitHub
GeoDCAT-AP alignment	
Issue (#370 derived from #82 GeoDCAT-AP) & #380	Relaxation approved
Issue #384	Relaxation approved
Issue #175 & #265 ; GeoDCAT-AP #68	Issue raised to GitHub

Feedback requests	
Issue #405	Issue raised to GitHub
Issue #348	Issue raised to GitHub
Issue #402	Issue raised to GitHub
Issue #386	Approved
Issue #404	Approved
Issue #295	Approved

Full Meeting Minutes

Welcome & Introduction Slides 1 - 7 Speaker: Pavlina Fragkou, Bert Van Nuffelen	PF welcomes the participants and goes over the webinar practicalities, followed by brief overview of SEMIC specifications including the Core Vocabularies and Application Profiles. Next PF introduces the objective of today's webinar. Address several key topics related to DCAT-AP: • Address the status of DCAT-AP: Overview of the outstanding issues and their resolution. • Specify the expected use of controlled vocabularies. • Discuss open issues related to the alignment with GeoDCAT-AP. • Introduce open issues for which community feedback is requested. • Handle changes in external vocabularies. BVN highlights the current status of DCAT-AP: • Total open issues: 43 ○ Bugs: 6 ○ Generic questions for information: 10 ○ Feedback requested: 7 • Potential releases for this year: ○ Bugfix release (Q2 2025): For issues that can be immediately fixed. ○ Minor release (Q4 2025): For issues that require further elaboration and have a significant impact on the community.
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Codelits Slides 10 - 37 Speaker: Bert Van Nuffelen	BVN mentions that the topic related to codelists was presented and subsequently discussed during the last GeoDCAT-AP webinar on 14th May 2024. However, it has not yet been addressed within the broader DCAT-AP community. The objective is to reach a consensus on a concrete proposal. The primary issue discussed in this section is the necessity to establish a common interpretation for the use of codelists in DCAT-AP. Specifically, the webinar addressed the interpretation of qualifications such as 'must,' 'is recommended,' and 'may' for codelists. The objective is to provide clear semantics for these qualifiers to ensure consistent implementation and validation across different profiles. Multiple systems scenario A scenario was presented to illustrate the challenges of using codelists across multiple systems: • Scenario: A publisher publishes successfully for System 1 but fails for System 2 due to different codelist requirements. • Solution: The webinar proposed the need for combined guidelines for publishers to ensure consistent validation across different systems Multiple systems scenario issue arises due to two interpretations for the mandatory codelist. BVN explained the two interpretations for mandatory codelists: • Interpretation 1: The value space is closed under the codelist, meaning only values from the codelist are allowed. ○ Strict validation: If a property has a value outside the codelist, it results in a validation violation. ○ Application: Conformant to the way literal value spaces are expressed, making it suitable for programming languages and most software systems. ○ Validation: Existing SHACL shapes can be used as-is, making validation simple and direct. Violations drive the feedback, ensuring strict adherence to the codelist. • Interpretation 2: At least one value from the codelist is required, allowing for additional values from other codelists ○ Application: where there is a need for aggregation at the level of properties, allowing for more flexibility. ○ Validation: Existing SHACL shapes cannot be used as-is and require a filtering process as part of the processing and conformance building. This filtering process is non-trivial to standardise and implement, making poor usages harder to detect. ▪ Matching SHACL shapes can be designed
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The choice between these two interpretations depends on the desired level of strictness and flexibility in the implementation of codelists.

Cross-profile assessment:

- Interpretation 1: More strict, requiring coordination among profiles to avoid conflicts. For example, if DCAT-AP uses Interpretation 2 and GeoDCAT-AP uses Interpretation 1, the usage in DCAT-AP would also need to adhere to Interpretation 1.
- Interpretation 2: Allows for more flexibility but still requires coordination to ensure consistent implementation across profiles.

Many participants believe that Interpretation 1 is harmful to interoperability and violates the open world assumption due to its limitation of a property to just one codelist. This view is widely supported, with many participants preferring Interpretation 2, as reflected in the general sentiment of the chat.

Issue [#407](#):

Necessity to confirm the current qualification of codelists use in DCAT-AP for various codelists.

SEMIC Proposition

Proposed usage for the codelists provided by SEMIC to be agreed on.

MUST	AT LEAST 1	RECOMMENDED	MAY
dcat:mediatype			dct:publisher
dcat:theme			dct:spatial
dct:accrualPeriodicity			
dct:format			
dct:language			
Adms:status			
dct:type (Agent Type)			
dct:type (licence type)			
dcatap:availability			
spdx:algorithm			
dct:accessRights			

Resolution

MP recommends usage of “may” for dct:accessRights instead of “must”. LR mentions usage of “recommended” for dct:accessRights given that they work on an application profile where they would like to use access right without a controlled vocabulary, LR also agrees for the other green propositions.

Additional question:

MC asked regarding the IANA codelist values are currently available only in XML format and not in RDF, in Controlled Vocabularies SHACL file stated that values can not be expressed in SHACL unless RDF format for IANA codelist is created. BVN suggests that, to align with

current RDF structural practices, the URLs of the IANA media type URIs should be tweaked and trusted to point to the correct concepts, even though they are not published as cost concepts. JK suggested moving this discussion to GitHub for further exploration and input for the best way to use IANA media types, keeping the issue open for now.

Issue #408:

The codelists “dct:type (Agent Type)” and “dct:type (licence type)” are currently interpreted as a ‘must’ in the qualification table. BVN raised concerns that the existing code lists for agents and licenses may not be comprehensive enough, or sufficiently centralised and harmonised to cover all necessary cases.

SEMIC proposition:

Change the qualification of the codelists “dct:type (Agent Type)” and “dct:type (licence type)” from “must” to “recommended”. It is still suggested to use these code lists, but not on a mandatory basis, allowing for more flexibility in their application.

MUST	AT LEAST 1	RECOMMENDED	MAY
dcat:mediatype		dct:type (Agent Type)	dct:publisher
dcat:theme		dct:type (licence type)	dct:spatial
dct:accrualPeriodicity			
dct:format			
dct:language			
Adms:status			
dct:type (Agent Type)			
dct:type (licence type)			
dcatap:availability			
spdx:algorithm	Motivation: codelist subject to local data portal interpretation		
dct:accessRights			

Resolution:

LR agreed on this proposal, as he raised the issue himself earlier. There were no objections from the participants. BVN emphasised that any comments should be directed to the relevant GitHub issue.

Issue #409:

BVN highlights the driving issue regarding the use of codelists “dcat:theme”. There are two formulations for the codelist “dcat:theme”: the individual usage note formulates the usage of “at least one” and the codelist section formulates the usage of “must,” which corresponds to a stricter interpretation.

The conflict affects the consistency and interoperability of “dcat:theme” and its associated codelists across different profiles. In practice many data portals have aligned with the EU NAL dataset theme vocabulary by extending it or mapping to it (e.g. INSPIRE)

SEMIC proposition:

The goal is to resolve this conflict and provide clear guidance on which interpretation to use. Three proposals are presented by BVN:

- Proposition A - MUST:

All other DCAT-AP profiles that would like to share their theme must introduce a subproperty of dcat:theme. dcat:theme stays paired exclusively with the EU NAL Dataset theme Vocabulary. Conflict stays with profiles beyond EU boundaries.
Bug fix release of the specification

- Proposition B – AT LEAST 1:
All data Catalogues must adapt their uniqueness expectation in the theme property and implement a filter based on the theme they like to show in the UI. The used codelists MUST be published as proper Linked Data in the public space. To facilitate every portal to make easy decisions.
Bug fix release of the specification.
- Proposition C – Conflict free resolution:
Replace the use of dcat:theme in DCAT-AP with a subproperty of dcat:theme. The new property r5r:theme stays paired exclusively with the EU NAL Dataset theme Vocabulary.
Minor release of the specification.

Resolution:

Priyanka mentioned that Proposition A is preferred because it allows for easier combination with other domains. She stated "A is preferred, otherwise you can't combine with other domains". When asked to elaborate, she explained that from a harvester's point of view, having a controlled vocabulary makes it easier to filter on certain domains. Without a controlled vocabulary, one would first have to figure out what all keywords relate to a specific domain, such as health. MP responded by sharing their experience with the Swedish data portal. They confirmed that this is a real problem and explained that they typically use faceting based on themes.

SVD addresses the concern that if the controlled vocabulary is limited to just one, it will hinder the adoption of other taxonomies. SVD agrees with MP's earlier point about using controlled vocabularies. SVD highlights the flexibility in expressing domain vocabularies and performing faceting and drill-down operations without being explicit, but notes that this introduces a runtime dependency. This flexibility allows for extending the system without writing new code and supporting multiple ways of faceting in a catalogue. SVD also mentions that fixing the system to one controlled vocabulary and requiring something not in DCAT seems counterintuitive. They suggest that it might be better to use the construct present in DCAT as intended and create a sub-property if something specific is needed, while expressing his preference for proposition B.

BVN remarks regarding proposition B it's crucial to publish and use code lists according to linked data principles. Failing to do so could lead to potential chaos. JK summarises from the chat that votes for B are slightly more favoured than for A. Therefore, further discussion on this issue will be moved to the GitHub.

**Issues with
GeoDCAT-AP
alignment**

[Slides 38 - 42](#)

Speaker: Bert Van
Nuffelen

Issue ([#370](#) derived from [#82](#) GeoDCAT-AP) & [#380](#):

The relaxation is regarding the cardinality of the property dct:rights. BVN explains that historically rights were limited to one document listing all rights. However, in GeoDCAT-AP, metadata often comes from translation of ISO to DCAT-AP, resulting in individual statements split into multiple lines and descriptions. This means there are multiple rights documents instead of just one. In DCAT-AP dct:rights is used for Catalogue and Distribution, with the current cardinality 0..1, in GeoDCAT-AP, the cardinality was relaxed to 0..*. In DCAT-AP HVD, dct:rights with cardinality 0..* is used also for Data Services.

SEMIC Proposition:

Given this history, it was proposed to increase cardinality. Relax cardinality for dct:rights to 0..* everywhere in DCAT-AP.

Resolution

There were no objections, and the relaxation is accepted.

Issue [#384](#):

BVN explained the conflicting cardinality of the spatial resolution with GeoDCAT-AP. There was a limitation in DCAT-AP where only one spatial resolution was available at the data set level. The community pointed out that spatial resolution for data sets should describe the resolutions in the original data, which might be captured in different resolutions.

SEMIC proposition:

Relax the DCAT-AP 3.0 cardinality of spatial resolution for Dataset to 0..*.

Resolution:

There were no objections, and the relaxation has been accepted.

Issue [#175](#) & [#265](#); GeoDCAT-AP [#68](#):

BVN emphasised the need for multiple representations of location information (WKT, GML, GeoJSON) due to varying formats.

Current SHACL constraints allow only one instance of locn:geometry, dcat:bbox, and dcat:centroid. However, GeoDCAT-AP permits multiple instances, representing one entity, described using different geometry encodings, e.g. WKT, GML and GeoJSON, leading to SHACL validation errors.

SEMIC proposition:

The current bounding box cardinality is 1, but multiple representations are needed, similar to multilingual texts, by lifting the cardinalities to 0..*.

Resolution:

MP suggested choosing a single format (WKT or GML) since GeoJSON has only one coordinate reference system. The issue of duplicate metadata was raised, and it was proposed that the community should harmonise on one format. JK highlighted that there was no consensus observed during the GeoDCAT-AP discussions on WKT or GML due to their different expressive powers. BVN agreed to take a note, as no immediate decision could be made, and community alignment was needed. JE agreed with MP's comment and additionally noted that in

	the final GeoDCAT-AP document, the metadata should be presented in a single, selected format.
Issues with a feedback request Slides 43 - 51 Speaker: Bert Van Nuffelen	Issue #405: BVN highlighted the need for a new property to provide more insight into the type of data service available. DCAT-AP provides no option for Data Service providers to classify the type of the Data Service. SEMIC proposition: Create subproperty dcatap:dataServiceType of dct:type to denote "types" of Data Services using a common codelist. Similar to the geodcatap:serviceType property introduced in GeoDCAT-AP with a codelist maintained by INSPIRE. Resolution: BVN suggested that the existing GeoDCAT-AP service type, maintained by the community, is very close to what is needed. SS mentioned that they had raised this issue and agreed that the proposition is a good idea, given that it is currently missing. MP debated whether a new property was necessary, suggesting that the existing DCT type could be used instead. The ambiguity remained and was further raised to GitHub. Issue #348: BVN mentioned the issue regarding identifying data set series in the catalog, as there is no direct property in DCAT for this purpose. How to include a dcat:DatasetSeries directly into a dcat:Catalog? In DCAT-AP 3.0.0, dcat:DatasetSeries is not a subclass of dcat:Dataset but of dcat:Resource, creating ambiguity in how to properly include it in a catalog. SEMIC proposition: • No support, wait for W3C (no release) • Introduce a separate r5r:datasetSeries property (equivalent to dcat:Dataset or dcat:DataService) (minor release) • Replace the use of dcat:Dataset and dcat:DataService with dcat:Resource and then it is up to the system to address the expectations. (minor release) Resolution: MP questioned the necessity of distinguishing between data sets and data set series, suggesting that the type property might complicate matters, MP proposed the usage of dcat:Dataset. The community needs to decide on the best approach to address this issue, with a call for participation on the GitHub issue to gather more viewpoints and collaboratively come to a concrete proposition for the autumn 2025 webinar. Issue #402: BVN highlighted the DCAT-AP use of SPDX for checksums to determine the status and quality of distributions. The release of SPDX 3.0 introduced significant changes from SPDX 2.0, including new URIs and a different structure. This change impacts DCAT-AP users who rely

	on SPDX 2.0 for checksum representation, as they would need to adapt to the new version without gaining additional benefits. SEMIC proposition: • Option A: replace SPDX 2.3 with SPDX 3.0 (minor release) • Option B: do not replace SPDX 2.3 but add usage note (bug fix release) Resolution: MC suggested to support both versions, allowing users to choose based on their system dependencies. However, this approach might complicate implementation for portal users. MP emphasised the need for a principle to handle dependencies on external vocabularies and the potential impact of such changes on the community. MP also mentioned the possible usage of proxy vocabularies in case there isn't commitment from the external vocabulary. The community needs to decide on the best approach to address this issue, with a call for participation on the GitHub issue. Issue #386: The current limitation in DCAT-AP, which allows only one publisher for a catalogue. Limitation problematic for catalogues maintained by multiple organisations SEMIC proposition: The proposal is the usage of foaf:Group instead if foaf:Agent for the range to bundle all members of the publisher group, without changing the cardinality Resolution: There were no comments provided by the participants. Issue #404: BVN highlighted the issue that DCAT-AP provides no option for data providers to make annotations regarding the intended target audience of a dataset. SEMIC proposition: Target audience is a vague concept. Objective is very portal usage dependent. Therefore, sharing it might be not informative. Proposal not to adopt this. Resolution: SS raised this issue because it relevance for more specific variants of DCAT-AP, such as in the health sector, where it might be important to distinguish between datasets for scientists and politicians. However, it was suggested that this should be handled in sector-specific DCAT-AP variants rather than in the central DCAT-AP. MG commented that it is not a good property to include in DCAT-AP due to the higher risk of misinterpretation. The general consensus was to not adopt the proposal. Issue #295: BVN raised an old issue again regarding the label of the property
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	change type seems not to match the definition and the mapping on the URI adms:status, before closing it due to the lack of comments. There were requests to provide a better definition or to rely on the community for feedback. However, no substantial feedback was received except for a recent mention to change the word "status" to "change type" in the definition. This suggestion was avoided to prevent a circular definition. SEMIC proposition: Since no feedback has been provided, proposal is to close the issue without consequence Resolution: There were no comments provided by the participants, agreement on the closing of this issue.
Next steps Slides 53 Speaker: Bert Van Nuffelen, Pavlina Fragkou	BVN wrapped up the session by summarising that small changes and relaxations will be included in a bug fix release, while bigger topics may require further proposals and discussions later in the year. A webinar will be held before the release to gather feedback and address smaller issues promptly. Participants are encouraged to provide additional feedback on open issues before April. BVN thanked everyone for their participation and interaction. PF expressed gratitude to the participants for their feedback, which are essential for making progress on the issues. PF also announced their departure from the role at SEMIC and mentioned that this was her last webinar. Additionally, PF extended special thanks to the team supporting the DCAT-AP, acknowledging their efforts.